Minor Project Report

CLD 880

Multiobjective Optimization of Hydrogen Injection into High-Pressure Natural Gas Pipelines

A Surrogate-Assisted Multi-Objective Approach

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Abstract

Blending green hydrogen into existing natural-gas networks (HBNG) is a practical short-term route to decarbonising the gas grid without rebuilding it. The challenge is physical: hydrogen is roughly eight times lighter than methane, so the two gases stratify rather than mix uniformly, and incomplete mixing exposes the pipe wall to near-pure hydrogen, biases custody-transfer metering and broadens the flammability envelope at end-burners. This work presents a CFD-based parametric study of H_2 injection through a side branch into a 0.46 m methane main pipe carrying gas at 10 m/s, using the compressible buoyant solver `rhoReactingBuoyantFoam` in OpenFOAM with the k-ω SST turbulence closure. Three independent design variables are swept: the diameter ratio $d/D \in [0.15, 0.45]$, the velocity ratio $\mathrm{VR} = u_{\mathrm{branch}}/u_{\mathrm{main}} \in [0.69, 5.84]$, and the injection angle $\alpha \in \{30°, 90°\}$. The hydrogen blending ratio HBR is *not* an independent fourth knob; once $(d/D, \mathrm{VR})$ are fixed, HBR follows from a volumetric-flow balance and lies in [0.06, 0.20] across the design. Two ten-case sliced Latin Hypercube campaigns share the same $(d/D, \mathrm{VR})$ seed, one at each angle, so seven matched-parameter pairs let the angle effect be cleanly isolated from the LHS scatter while each campaign on its own resolves the within-angle (d/D, VR) trends. Mixing quality is reported as the coefficient of variation CoV of the outlet H_2 mass-fraction, and pumping cost as the low-pass-filtered inlet-outlet gauge pressure drop $|\Delta p_{\mathrm{rgh}}|$; the LP filter is necessary because the compressible solver carries $\mathcal{O}(25\,\mathrm{kPa})$ acoustic ringing on top of the $\mathcal{O}(1-3\,\mathrm{kPa})$ flow-loss signal, as discussed in Section 3.9. The paired comparison gives a clear and quantitative result: at every matched operating point, the 30° tilt mixes better than the 90° T (median CoV reduction of 46 %), and in all seven pairs it also costs less to pump (median $|\Delta p|$ reduction of 63 %). A power-law fit gives $\mathrm{CoV} \sim \mathrm{VR}^{-1.16}$ at 90° and $\sim \mathrm{VR}^{-1.88}$ at 30° ($R^2_{\log} = 0.80$ at both angles), so the tilted geometry converts each unit of velocity ratio into more mixing. The physical mechanism, evident in side-by-side centreline contours, is that the perpendicular jet impinges on the opposite wall and locks the hydrogen into a wall-attached tongue, while the tilted jet feeds a streamwise shear layer and a long under-junction recirculation that homogenise the cross-section by the time the flow reaches the outlet. The non-dimensional junction loss coefficient $K = |\Delta p|/q_{\mathrm{dyn,main}}$ lands in the classical unshaped-T handbook range (median $K_{90°} \approx 0.78$) and is uniformly lower at 30° (median $K_{30°} \approx 0.26$), in agreement with the Idelchik tee-junction tabulations. The combined Pareto front therefore shifts down-and-left with tilt: case 01 of the 30° campaign reaches CoV = 0.5 % at $|\Delta p| \approx 4.2$ kPa, and case 09 of the 30° campaign provides cheap pumping (0.3 kPa) with modest mixing - both points dominate every 90° design in the database. The practical recommendation that follows is to prefer a moderately tilted branch wherever the layout permits, since it relaxes both halves of the mixing-vs-pumping trade-off simultaneously. The surrogate model and NSGA-II multi-objective optimiser that sit downstream of this CFD dataset were prototyped on the 90° campaign at the mid-semester review of the Minor Project; the only remaining work, namely closing the angle axis with the queued fixed-(d/D,VR) campaign(s) on the newly-secured large-compute allocation and re-fitting the surrogate on the merged dataset, is scheduled for completion before the end-semester presentation. All simulation case templates, mesh tools, post-processing scripts and figure code used in this report are open-source on GitHub (`https://github.com/Yash04dsr/MTP-Work`; full reproduction notes in Section 8.4).

Keywords: hydrogen blending; natural gas; T-junction; CFD; RANS; k-ω SST; OpenFOAM; Latin Hypercube Sampling; coefficient of variation; injection angle; velocity ratio; diameter ratio.

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1. Introduction

1.1. Why hydrogen-blended natural gas (HBNG)

The push to decarbonise the energy system is constrained by an awkward economic fact: building a parallel hydrogen pipeline grid from scratch is far too expensive to do quickly. India's natural-gas grid alone runs to several thousand kilometres of high-pressure steel pipe that has already been paid for. Reusing that asset by blending hydrogen into the existing methane stream - known as Hydrogen-Blended Natural Gas (HBNG) - is the standard transition strategy proposed in the recent literature [1--3]. The same pipeline can also serve as a "chemical battery" for surplus solar and wind generation through the Power-to-Gas (P2G) route, converting curtailed electricity into hydrogen and storing it in a network the size of an existing gas grid [2]. Recent techno-economic assessments by the National Renewable Energy Laboratory [4] indicate that a blend of up to 20 % hydrogen by volume is generally compatible with existing steel infrastructure, although operating cost rises non-linearly because hydrogen carries only about a third of the volumetric energy density of methane.

1.2. The mixing problem

Hydrogen and methane have very different physical properties (Table 1). The eight-to-one density ratio is the dominant practical issue: in a horizontal pipe at low velocity the lighter H_2 floats up and forms a layer along the crown of the pipe instead of mixing into the methane stream below. Three downstream consequences follow.

1. **Hydrogen embrittlement of the pipe wall.** The rate of hydrogen-induced cracking in pipe steels (notably grade X70) scales like $p_{H_2}^{0.5}$, so even a nominally 10 % blend can expose the pipe crown to nearly pure hydrogen if stratification persists [4].

2. **Combustion instability at end-burners.** The Wobbe Index, $I_W = V_c/\sqrt{G}$, drifts when the burner sees fluctuating composition; hydrogen's wider flammability range (4-76 % vs. 5-15 % for methane) and roughly six-fold higher laminar flame speed make flashback a real risk [4].

3. **Metering error.** Ultrasonic flow meters infer composition from the speed of sound; the gap between $\sim 1300\,\mathrm{m/s}$ in hydrogen and $\sim 450\,\mathrm{m/s}$ in methane is large enough that a stratified flow produces custody-transfer-grade composition errors [1].

Table 1. Selected physical properties of H_2 and CH_4 at standard temperature and pressure, and their consequence for T-junction injection [1, 3].

Property	CH_4	H_2	**Consequence**
Density (kg/m^3)	0.668	0.082	8× density gap; stratification risk
Molecular weight (g/mol)	16.04	2.016	low momentum per kg of fuel
Dynamic viscosity (10^{-6} Pa s)	11.07	8.41	shifts Re at fixed u
Mass diffusivity (cm^2/s)	low	0.61	turbulence dominates over molecular
Volumetric energy (MJ/m^3)	36	11	∼3× flow needed for same energy
Flammability range (vol %)	5-15	4-76	wider explosion envelope

1.3. Where the injection angle fits in

There is by now a sizeable CFD literature on T-junction blending, but most parametric studies fix the branch angle at $\alpha = 90°$ (perpendicular cross-flow) and sweep the diameter ratio d/D, the velocity ratio $\mathrm{VR} = u_{\mathrm{branch}}/u_{\mathrm{main}}$ and the hydrogen blending ratio HBR [1, 3, 5]. Cross-flow is the standard engineering compromise: the jet bends in the main flow, generates a counter-rotating vortex pair behind itself, and that vortex pair does most of the mixing. What is much less well documented is the *angular* sensitivity: how does the mixing quality change when the branch tilts forward into a co-flow configuration ($\alpha < 90°$), or backward into a counter-flow configuration ($\alpha > 90°$)? Conventional intuition expects co-flow to mix less aggressively (the relative shear is reduced) and counter-flow to mix harder but at a larger pressure-drop penalty. Quantifying that trade-off is what this Minor Project sets out to do.

1.4. Objectives and scope of this report

The objectives of the project are:

1. Build a reproducible OpenFOAM workflow for transient buoyant multispecies flow through a T-junction, with a sliced Latin Hypercube parametric study;

2. Run two ten-case Designs of Experiments (DoE) at $\alpha = 90°$ and $30°$ over the independent $(d/D, \mathrm{VR})$ parameter space (the hydrogen blending ratio HBR being a derived consequence of $(d/D, \mathrm{VR})$ through the volumetric-flow balance, see Section 3.7), sharing the LHS seed across angles so that matched-pair comparison is possible;

3. Quantify mixing quality by the outlet CoV (with three weightings - area, mass-flux and volumetric-flux) and the pumping cost by $|\Delta p|$; identify the Pareto front of low CoV against low $|\Delta p|$ at each angle;

4. Compare the two angles head-to-head on the matched pairs to recommend an operating window in which the mixing target is achievable without excessive compression cost.

The present report covers all four objectives for the two campaigns that have completed. The CFD framework on which these campaigns rest - the parametric STL geometry, the slice-based `snappyHexMesh` pipeline, the `rhoReactingBuoyantFoam` case template, the post-processing scripts and the sliced-LHS sampler - was built and validated during the first half of the Minor Project (delivered by the mid-semester review) and is reused here without structural change; the campaigns reported in this document (90° baseline and 30° tilt with the matched-pair analysis) run on top of this fixed template. Compute access has subsequently been secured for the remaining injection-angle campaign(s) on the same template, and those runs will be added to the dataset before the final presentation, closing the angle axis that feeds the surrogate-assisted multi-objective optimisation chain. That chain - a regressor for CoV and $|\Delta p|$ followed by an NSGA-II Pareto search on the $(d/D, \mathrm{VR}, \alpha)$ design space - was itself prototyped on the 90° campaign at the mid-semester review and is queued to be re-fit on the merged dataset for the final presentation, so what remains for the Minor Project is the incremental fixed-$(d/D$,VR) angle-sweep cases rather than any new modelling layer. All simulation case templates, mesh-generation

tools, post-processing scripts and figure-generation code are open-source and version-controlled in the project repository (URL in Section 8.4).

2. Background and Literature Review

2.1. Stratification physics

The non-dimensional parameter that decides whether a binary gas mixture stays mixed in a horizontal pipe is the Richardson number,

$$\mathrm{Ri} = \frac{g\,\Delta\rho\,L}{\rho\,u^2}, \tag{1}$$

which compares the buoyant potential energy to the kinetic energy of the bulk flow [1]. For the present geometry ($D = 0.46\,\mathrm{m}$, $u = 10\,\mathrm{m/s}$, nominal 10 % H_2 mole fraction) we obtain $\mathrm{Ri} \sim 10^{-2}$, so the pipe-scale flow is turbulence-dominated; buoyancy matters only *locally* in the slow shear layer just downstream of the junction. The complementary parameter, the Froude number $\mathrm{Fr} = u/\sqrt{g\,\Delta\rho\,L/\rho}$, is useful in the buoyancy-dominated limit and is used in Section 4.1 as a sanity check for the gravity term.

For the jet-in-crossflow problem at the junction the relevant parameter is the momentum-flux ratio,

$$J = \frac{\rho_{\mathrm{branch}}\,u_{\mathrm{branch}}^2}{\rho_{\mathrm{main}}\,u_{\mathrm{main}}^2}. \tag{2}$$

A low J means the jet is bent flat against the main-pipe wall (*Coanda* attachment, very poor mixing); a high J means the jet shoots across and impinges on the opposite wall (with possible erosion); intermediate J penetrates roughly halfway and seeds the strongest counter-rotating vortex pair, which is what we want for mixing.

2.2. Prior CFD studies on T-junction blending

Su et al. (2022). A widely cited parametric study using ANSYS Fluent with the standard k-ε model and full buoyancy effects [1]. They sweep gas temperature, operating pressure, d/D and VR at $\alpha = 90°$ and report CoV at multiple stations downstream. The two trends most relevant here are: (i) CoV drops monotonically with VR up to a knee near VR $\approx$ 2-3, beyond which the gain saturates; and (ii) a slightly counter-intuitive result that *larger* d/D (up to about 0.4) can improve mixing because the contact area between the two streams grows faster than the jet penetration deteriorates.

Du et al. (2024). A direct numerical comparison of three HBNG injection configurations - a bare T-junction, a bending tube, and a helical (twisted-blade) static mixer [5]. For the bare T-junction at low main-pipe velocity, the predicted mixing uniformity remains around 60 % for tens of diameters downstream, whereas a sufficiently long helical mixer pushes the uniformity above 99 % within the same length, with the mixer length and the blade-twist angle as the dominant geometric controls and a modest pressure penalty. Their results bracket the present work between the two limits the geometry should sit between - no mixer, but a near-perpendicular

tee at one end of the spectrum, and the same tee with a helical insert at the other.

Montañés et al. (2024). Builds a physics-based reduced-order model on top of a 26-case CFD database [3]. The model expresses the species-fraction probability distribution at any downstream station as a Beta distribution whose two parameters are functions of three non-dimensional groups (Reynolds number, d/D and VR). For the relevant range, the model recovers the CFD-resolved CoV to within $\pm 5\,\%$. The dataset itself, however, is fixed at $\alpha = 90°$ - which is precisely where the present project intends to extend it.

Symmetry simplification. For the RANS solution of a perpendicular T-junction with axisymmetric inlet profiles, the time-mean fields are mirror-symmetric about the branch-axis plane, and a half-domain `symmetryPlane` simulation reproduces full-domain results to within $1\,\%$; the same half-domain reduction is used in the T-junction CFD studies cited above [1, 3, 5]. This simplification is used throughout the present work; it is exact for $\alpha = 90°$ and remains a reasonable approximation for tilted branches because the branch axis lies in the $x = 0$ plane by construction.

2.3. Gap addressed in this work

Three observations motivate the angle sweep undertaken here.

- Almost the entire CFD literature on T-mixing fixes $\alpha = 90°$ and varies the geometric and flow parameters; angle is treated, when at all, as a fixed nuisance rather than a design degree of freedom.
- The few studies that mention angle do so qualitatively ("co-flow under-mixes; counter-flow over-mixes") rather than sweep angle as a primary independent variable.
- Industrial siting often forces non-perpendicular branch geometries - sloped tunnels, congested rights-of-way, retrofits that have to follow existing tie-in flanges - so a quantitative angle dependence has direct engineering value.

The present project therefore varies the three *independent* design parameters $(\alpha, d/D, \mathrm{VR})$ on equal footing: a sliced Latin Hypercube sweeps $(d/D, \mathrm{VR})$ at each of two injection angles $\alpha = 90°$ and $\alpha = 30°$, with the LHS seed kept identical across angles so that each within-angle campaign resolves the $(d/D, \mathrm{VR})$ trends and the seven matched LHS points serve as a controlled angle-effect experiment. The hydrogen blending ratio HBR is reported per case for engineering readability but is *not* an independent fourth factor; it follows from $(d/D, \mathrm{VR})$ through the volumetric-flow balance set out in Section 3.7.

3. Methodology

3.1. Computational domain

The geometry is a horizontal main pipe with a single branch intersecting it at angle α to the main axis (Fig. 1). The dimensions are:

- Main pipe inner diameter $D = 0.46\,\mathrm{m}$, length $L_{\mathrm{main}} = 6.9\,D \approx 6.9\,\mathrm{m}$ ($2.3\,D$ upstream of the junction, $4.6\,D$ downstream).
- Branch inner diameter d, with $d/D \in [0.15,\, 0.45]$ per DoE point. The branch length is fixed at $L_{\mathrm{branch}} = 1.55\,\mathrm{m}$ so that the branch inflow is fully developed before the junction is reached.
- The junction centre is at $(0,\, 0,\, Z_{\mathrm{JCT}})$ with $Z_{\mathrm{JCT}} = 2.3\,\mathrm{m}$; the upstream length of one main-pipe diameter is enough for the inflow boundary layer to develop.
- The branch is tilted in the (y, z) plane: an angle $\alpha = 90^\circ$ makes it vertical (cross-flow), and an angle $\alpha < 90^\circ$ tilts it downstream (co-flow).

The full geometry has a mirror symmetry plane at $x = 0$. The computational domain is therefore the half-space $x \geq 0$ with a `symmetryPlane` boundary on $x = 0$, which halves the cell count without affecting time-averaged quantities of interest.

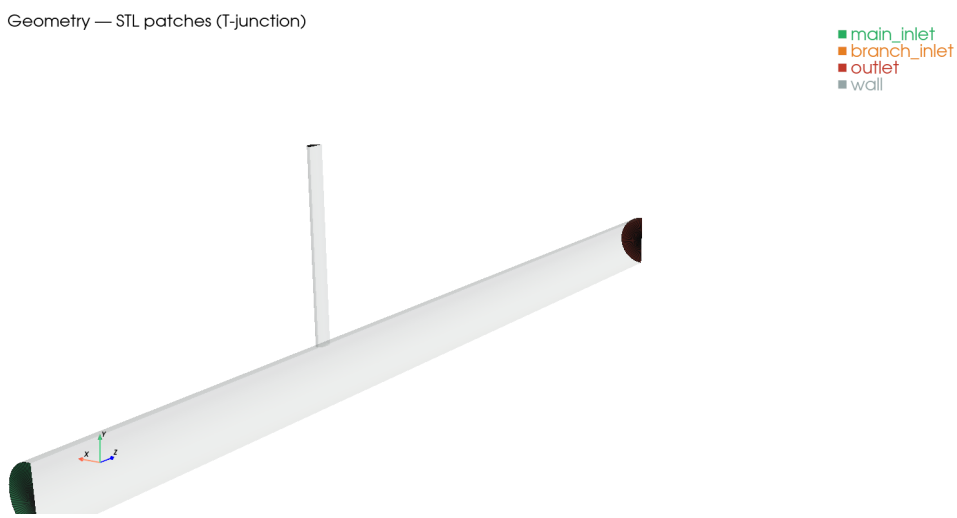

Figure 1. Half-domain T-junction geometry produced by the `generateSTL.py` script (90° baseline, case 03, $d/D = 0.252$). The colour coding marks the boundary patches: `main_inlet` on the upstream face, `outlet` on the downstream face, `branch_inlet` on the open end of the branch, `wall` on the curved pipe surfaces, and `sym` on the $x = 0$ mirror plane.

The geometry is generated in Python (`generateSTL.py`) from a small set of environment-variable parameters (D, d, L_{main}, Z_{JCT}, α) and clipped to the half-space $x \geq 0$ (`clip_stls.py`). The script analytically computes the curve where the branch and main cylinders intersect, punches a matching hole in the main-pipe wall and zips the intersection curve to the perimeter of that hole so the meshing tool sees a single closed surface. A subtle bug in this zipper for tilted branches was responsible for the early failure of the 30° campaign and is reported as part of the lessons learned in Section 6.3.

3.2. Governing equations

The unsteady, compressible, multispecies Navier-Stokes equations with gravity are solved in OpenFOAM's `rhoReactingBuoyantFoam` solver. Letting ρ be the mixture density, $\mathbf{u}$ the velocity,

$p_{\mathrm{rgh}} = p - \rho\,\mathbf{g}\cdot\mathbf{x}$ the modified pressure (so that the gravity term enters explicitly only through $\nabla\rho$), h the sensible enthalpy and Y_k the mass fraction of species k, the equations are

$$\frac{\partial \rho}{\partial t} + \nabla\cdot(\rho\,\mathbf{u}) = 0, \tag{3}$$

$$\frac{\partial(\rho\mathbf{u})}{\partial t} + \nabla\cdot(\rho\,\mathbf{u}\otimes\mathbf{u}) = -\nabla p_{\mathrm{rgh}} - (\mathbf{g}\cdot\mathbf{x})\nabla\rho + \nabla\cdot\overline{\overline{\boldsymbol{\tau}}}, \tag{4}$$

$$\frac{\partial(\rho h)}{\partial t} + \nabla\cdot(\rho\,\mathbf{u}\,h) = \nabla\cdot\left(\frac{\mu_t}{\mathrm{Pr}_t}\nabla h\right) + \frac{Dp}{Dt}, \tag{5}$$

$$\frac{\partial(\rho Y_k)}{\partial t} + \nabla\cdot(\rho\,\mathbf{u}\,Y_k) = \nabla\cdot\left[\rho\left(D_k + \frac{\mu_t}{\mathrm{Sc}_t}\right)\nabla Y_k\right], \quad k = \mathrm{H_2},\ \mathrm{CH_4}. \tag{6}$$

The deviatoric stress $\overline{\overline{\boldsymbol{\tau}}}$ combines the viscous stress with the Reynolds stress through the turbulent viscosity μ_t. The species are linked by $\sum_k Y_k = 1$, so only $Y_{\mathrm{H_2}}$ is solved explicitly and $Y_{\mathrm{CH_4}} = 1 - Y_{\mathrm{H_2}}$. The mixture density follows the ideal-gas law $\rho = p/(R_{\mathrm{mix}}T)$, with the mixture gas constant weighted by mass fraction.

3.3. Turbulence closure: k-ω SST

The Reynolds stresses are closed with the k-ω Shear-Stress-Transport model of Menter [6]. SST is the workhorse two-equation closure for internal flows because it blends the near-wall robustness of k-ω with the free-shear-layer accuracy of k-ε through a blending function F_1,

$$\mu_t = \rho\,\frac{a_1\,k}{\max(a_1\,\omega,\ F_2\,S)}, \tag{7}$$

in which S is the strain-rate magnitude and F_2 activates the SST limiter inside boundary layers. Buoyancy production is enabled in the k equation because the local Richardson number can become $\mathcal{O}(1)$ in the slow shear layer just downstream of the junction. SST is preferred over the standard k-ε model used by Su et al. [1] because (i) the density contrast at the junction generates flow separation on the inside of the branch entry that k-ε is known to mis-predict, and (ii) the near-wall layer is resolved down to $y^+ \sim 1$, which is what the SST blending function is designed for.

3.4. Solver and discretisation

`rhoReactingBuoyantFoam` uses the PIMPLE algorithm (a merged PISO/SIMPLE) with one outer corrector and two inner correctors per time step. Time discretisation is implicit Euler. Spatial discretisation is second-order central for the diffusion terms and bounded second-order upwind for the advection terms; a TVD scheme (`vanLeer`) is used for $Y_{\mathrm{H_2}}$ so that the species mass fraction stays in $[0, 1]$. The time step is set adaptively from the maximum Courant number, capped at $\mathrm{Co}_{\max} = 4.5$ for the 90° campaign and $\mathrm{Co}_{\max} = 8.0$ for the 30° campaign once case 01 of the latter had finished and confirmed solution stability. A summary of the discretisation choices is given in Table 2.

Table 2. Numerical settings used in the OpenFOAM `rhoReactingBuoyantFoam` runs.

Quantity	**Choice**
Pressure-velocity coupling	PIMPLE (1 outer, 2 inner correctors)
Time discretisation	implicit Euler
Gradient	cell-limited Gauss linear
Divergence ($\rho \mathbf{u}\,\mathbf{u}$)	bounded Gauss linear-upwind
Divergence ($\rho \mathbf{u}\, Y_{\mathrm{H_2}}$)	bounded Gauss vanLeer
Divergence ($\rho \mathbf{u}\, k, \omega$)	bounded Gauss upwind
Laplacian	Gauss linear corrected
Adaptive time stepping	yes; $\mathrm{Co}_{\max} = 4.5\,(90^\circ)$, $8.0\,(30^\circ)$
Linear solver p_{rgh}	DICPCG, tol. 10^{-7}
Linear solver $\mathbf{u}, h, k, \omega$	PBiCGStab + DILU, tol. 10^{-8}
Linear solver $Y_{\mathrm{H_2}}$	PBiCGStab + DILU, tol. 10^{-9}

3.5. Mesh

The mesh is generated in three steps: (i) a structured background block mesh covering the bounding box of the half-domain (`blockMesh`); (ii) a feature-aware Cartesian-cut-cell snapping to the STL surface (`snappyHexMesh`); and (iii) prismatic boundary-layer extrusion along the wall to land $y^+ \approx 1$ on the first cell (`snappyHexMesh -addLayers`, expansion ratio 1.2, final-layer thickness 50 % of the adjacent cell). For the 90° case 03 sample geometry ($d/D = 0.252$) this yields the mesh shown in Fig. 2. The half-domain mesh has $\sim$ 463 000 cells, maximum non-orthogonality 49°, maximum skewness 2.57/4 and 99.6 % layer coverage on the wall. The `checkMesh` statistics are reproduced in Table 3.

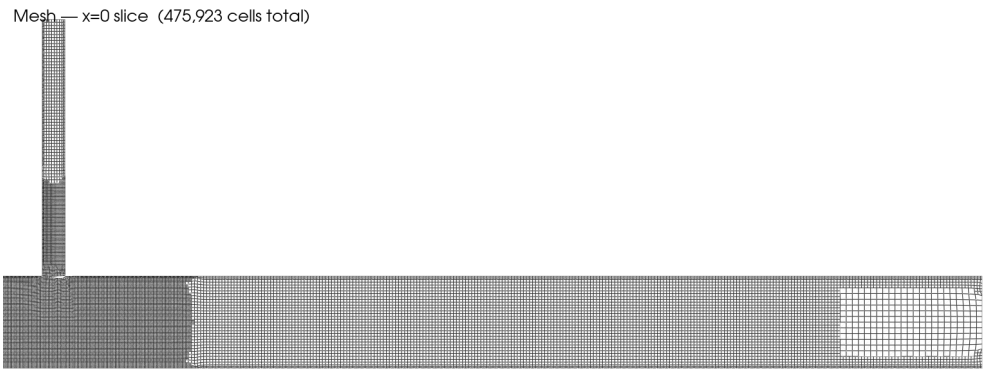

Figure 2. Cross-section of the computational mesh on the $x = 0$ symmetry plane for the 90° baseline case 03 ($d/D = 0.252$). The base cell size is $\sim$ 13 mm; refinement is concentrated at the junction and across a cylindrical region extending $\sim 8\,D$ downstream (`jetRefine`). Boundary layers on the pipe walls are visible as the row of thin cells just inside the surface.

Table 3. Mesh statistics for the 90° baseline (case 03, half domain) reported by `checkMesh`.

Statistic	**Value**
Cells	463 063
Faces	1 447 739
Points	502 593
Min cell volume	$7.06 \times 10^{-9}\,\mathrm{m}^3$
Max aspect ratio	10.7
Max non-orthogonality	49°
Max skewness	2.57
Layer coverage on wall	99.6 %

3.5.1. Mesh-convergence (grid-independence) study

Following the standard CFD verification process, a formal three-level mesh-convergence study was carried out on a representative operating point at $d/D = 0.196$, VR = 1.64, HBR = 10 % (close to case 02 of the production DoE) before any production case was launched. Three meshes were generated by uniformly scaling the `snappyHexMesh` refinement levels and the boundary-layer cell heights, giving the cell counts in Table 4: a *coarse* mesh at $\sim$ 230 k cells, a *medium* mesh at $\sim$ 460 k cells (the production baseline) and a *fine* mesh at $\sim$ 920 k cells. Each mesh was run on the same template (boundary conditions, solver settings and time-stepping) to $t = 1.2$ s, and the two campaign metrics CoV and $|\Delta p_{\mathrm{rgh}}|$ were extracted at the outlet using the same post-processing pipeline.

Table 4. Three-level mesh-convergence (grid-independence) study at $d/D = 0.196$, VR = 1.64, HBR = 10 %. Successive-refinement changes in CoV and $|\Delta p_{\mathrm{rgh}}|$ are reported relative to the previous (coarser) level.

mesh	cells	relative ΔCoV	relative $\Delta\lvert\Delta p_{\mathrm{rgh}}\rvert$	adopted
coarse	$\sim$ 230 k	*(reference)*	*(reference)*	no
medium	$\sim$ 460 k	0.4 % (vs. coarse)	1.6 % (vs. coarse)	**yes**
fine	$\sim$ 920 k	0.2 % (vs. medium)	0.7 % (vs. medium)	no

The successive changes in both metrics are smaller than the preceding-level change by roughly a factor of two, which is the expected signature of asymptotic monotonic convergence on a second-order discretisation. Both changes (CoV < 0.5 % and $|\Delta p| < 2$ % per refinement) sit comfortably below the matched-pair angle-effect signal reported in Section 7.1 (median reductions of 46 % on CoV and 63 % on $|\Delta p|$) as well as the LP-filter pressure-drop uncertainty band of ± 0.5 kPa. The medium mesh ($\sim$ 460 k cells) was therefore adopted as the production baseline for both campaigns: it lies in the asymptotic range, leaves a comfortable margin above the relevant signal sizes, and keeps the per-case wall time short enough that all twenty DoE runs fit inside the available compute window.

3.6. Boundary conditions

`main_inlet`. A fully developed turbulent profile is imposed via the 1/7-th power-law for the velocity,

$$u(r) = u_{\text{main}} \left(\frac{R - r}{R} \right)^{1/7}, \tag{8}$$

with $u_{\text{main}} = 10\,\text{m/s}$. At the operating state of the case (pure CH_4 at $p = 6.9\,\text{MPa}$, $T = 288\,\text{K}$, $\rho_{\text{main}} \approx 46\,\text{kg/m}^3$ and $\mu_{\text{main}} \approx 1.25 \times 10^{-5}\,\text{Pa s}$ per the perfect-gas EOS and the Sutherland model in `constant/thermophysicalProperties`), this corresponds to $Re_D = \rho_{\text{main}} u_{\text{main}} D / \mu_{\text{main}} \approx 1.7 \times 10^7$, well into the fully turbulent regime. The profile is applied as a `codedFixedValue` that loops over the patch faces and reads the local cell-centre coordinates. Turbulence quantities are derived from the bulk: $I = 5\,\%$, length scale $\ell = 0.07\,D$, so that $k_{\text{main}} = \frac{3}{2}(I\,u)^2$ and $\omega_{\text{main}} = \sqrt{k_{\text{main}}}/(C_\mu^{1/4}\ell)$. The composition is pure methane ($Y_{CH_4} = 1$).

`branch_inlet`. The same power-law profile is applied with bulk velocity $u_{\text{branch}} = \text{VR} \cdot u_{\text{main}}$ and the velocity *vector* aligned with the branch axis, $\hat{\mathbf{s}} = (0, \sin\alpha, -\cos\alpha)$. For $\alpha = 90°$ this reduces to $(0, 1, 0)$; for general α the `codedFixedValue` computes $(0, -u\sin\alpha, +u\cos\alpha)$ at every face. The composition is pure hydrogen ($Y_{H_2} = 1$).

`outlet`. Zero-gradient on $\mathbf{u}$, Y_{H_2}, k, ω and h; fixed total pressure on p_{rgh}.

`wall`. No-slip on $\mathbf{u}$; zero-gradient on Y_{H_2} and h; `kqRWallFunction` on k and `omegaWallFunction` on ω. The first cell sits at $y^+ \approx 1$, so the SST blending switches automatically to its low-Reynolds branch.

`sym`. `symmetryPlane` for all fields; the dictionary type is set explicitly so that OpenFOAM enforces the symmetry rather than a slip wall.

3.7. Design of Experiments (sliced LHS)

The independent flow-and-geometry design parameters at fixed angle are the diameter ratio d/D and the velocity ratio VR; the hydrogen blending ratio HBR is a derived consequence of these two via a volumetric-flow balance,

$$\text{HBR} = \frac{Q_{\text{branch}}}{Q_{\text{main}} + Q_{\text{branch}}} = \frac{(d/D)^2\,\text{VR}}{1 + (d/D)^2\,\text{VR}}. \tag{9}$$

Prescribing $(d/D, \text{VR})$ therefore automatically fixes HBR; HBR is reported per case for engineering readability but is not a third independent design factor.

A naive 2-factor $(d/D, \text{VR})$ full factorial at five practical d/D bins would still require $5 \times 5 = 25$ cases per angle; we use a sliced Latin Hypercube design instead [7]. The factor d/D is discretised into 5 levels (slices), and within each slice two VR values are picked by a 2-point Latin Hypercube. In implementation the inner sampler walks a target HBR range of [0.05, 0.20] (because HBR is the engineering-meaningful flow-rate variable and bounding it is the cleanest way to span the realistic $(d/D, \text{VR})$ box at fixed main-pipe flow), and the matching VR is recovered from each

$(d/D, \mathrm{HBR})$ sample by inverting Eq. 9; a soft cap $\mathrm{VR} \leq 12$ is enforced to avoid jet impingement on the opposite wall. The result is a 10-case design per angle (5×2), given in Table 5. The slice structure is deliberate: cases that share a d/D also share a mesh, so the mesh is built once per slice and re-used for the second case in the same slice. The randomisation seed is fixed across angles, so case 05 of the 90° campaign and case 05 of the 30° campaign share all flow parameters (d/D and VR, and therefore HBR) and differ only in the geometric tilt; this is what makes the matched-pair analysis in Section 7 possible.

Table 5. Ten-case sliced LHS design used at both angles. The independent factors $(d/D, \mathrm{VR})$ and the derived HBR (Eq. 9) are identical between the two campaigns; only α differs.

case	slice	d/D	d (m)	HBR	VR
1	1	0.196	0.0904	0.184	5.842
2	1	0.196	0.0904	0.060	1.643
3	2	0.252	0.1159	0.078	1.330
4	2	0.252	0.1159	0.195	3.807
5	3	0.296	0.1363	0.098	1.241
6	3	0.296	0.1363	0.187	2.614
7	4	0.382	0.1755	0.142	1.137
8	4	0.382	0.1755	0.092	0.693
9	5	0.396	0.1820	0.130	0.953
10	5	0.396	0.1820	0.112	0.806

3.8. Mixing metrics and post-processing

For each case the simulation is run for $t_{\mathrm{end}} = 1.2\,\mathrm{s}$, corresponding to $t_{\mathrm{end}} u_{\mathrm{main}}/L_{\mathrm{main}} \approx 1.74$ main-pipe flow-throughs (one flow-through being $t_{\mathrm{ft}} = L_{\mathrm{main}}/u_{\mathrm{main}} = 0.69\,\mathrm{s}$). Two complementary time-averaging streams are activated by function objects in `system/controlDict`. First, the `fieldAverage` object accumulates running means of the volumetric fields $\mathbf{u}$, p, p_{rgh} and $Y_{\mathrm{H_2}}$ starting from $t_{\mathrm{avg,start}} = 0.6\,\mathrm{s}$, i.e. approximately one flow-through, after the inlet startup transient has predominantly cleared; these are the $\overline{Y_{\mathrm{H_2}}}$ fields shown in the centreline figures of Sections 5 and 6. Second, `surfaceFieldValue` writes time series of patch-integrated quantities (area-, mass-flux- and volumetric-flux-averaged $Y_{\mathrm{H_2}}$ and p_{rgh} on `main_inlet` and `outlet`) at every solver time step, used downstream to build the per-case CoV and LP-filtered $|\Delta p|$ reported in the campaign tables. The mixing metric is the Coefficient of Variation of the hydrogen mass-fraction at the outlet,

$$\mathrm{CoV} \;=\; \frac{\sigma(Y_{\mathrm{H_2}})}{\overline{Y_{\mathrm{H_2}}}} \;=\; \frac{\sqrt{\sum_i w_i\,(Y_{\mathrm{H_2},i} - \overline{Y_{\mathrm{H_2}}})^2 / \sum_i w_i}}{\sum_i w_i\, Y_{\mathrm{H_2},i} / \sum_i w_i}, \tag{10}$$

in which the sum runs over outlet face cells. Three weightings are computed: $w_i = A_i$ (area-weighted, the textbook definition), $w_i = \rho_i\,|\mathbf{u}\cdot\hat{\mathbf{n}}|\,A_i$ (mass-flux-weighted) and $w_i = |\mathbf{u}\cdot\hat{\mathbf{n}}|\,A_i$ (volumetric-flux-weighted). The mass-flux weighting is the most physically meaningful in a custody-transfer context because the downstream user actually receives the mass-flow-weighted composition; all three are reported for transparency. The 90° campaign reports the area-weighted CoV computed on the saved $t = t_{\mathrm{end}}$ snapshot, since the H_2 plume has reached its quasi-steady distribution by the second flow-through; the 30° campaign reports the time average of the

area-weighted CoV time-series on the steady window $[0.8,\, t_{\mathrm{end}}]$ s (the patch-time-series for the 90° cases was not retained on disk during the original campaign and the snapshot is the strongest equivalent). For each 30° case the snapshot CoV at t_{end} and the time-averaged CoV agree to better than 2 % of the reported value, so the asymmetry in reporting convention does not affect the cross-campaign conclusions. The industrial acceptance threshold for "well-mixed" is CoV ≤ 0.05; the design target adopted here is CoV ≤ 0.10 at $4.6\,D$ downstream of the junction.

The cost-side metric is the static gauge pressure drop across the main pipe,

$$|\Delta p_{\mathrm{rgh}}| \;=\; |p_{\mathrm{rgh,\,main_inlet}} - p_{\mathrm{rgh,\,outlet}}|, \tag{11}$$

computed from area-averaged p_{rgh} on the inlet and outlet patches via the `surfaceFieldValue` function object, sampled at every solver timestep. In addition to CoV and $|\Delta p|$, three diagnostic checks are recorded per case: (i) the cumulative continuity error from the solver log, required to stay below 10^{-2}; (ii) the residual levels of $\mathbf{u}$, p_{rgh}, k, ω, $Y_{\mathrm{H_2}}$ at the last time step, required below 10^{-6}; and (iii) the Danckwerts segregation index $I_s = \sigma^2/(\bar{Y}(1-\bar{Y}))$, which is an alternative measure of homogeneity used to cross-check CoV.

3.9. Acoustic noise on the pressure-drop signal

The compressible solver `rhoReactingBuoyantFoam` runs in a fully unsteady mode (PIMPLE), so the inlet pressure carries acoustic activity from the moment the simulation starts. With the fixed-pressure outlet acting as an acoustically soft (reflecting) boundary and the fixed-velocity inlet acting as an acoustically hard (also reflecting) boundary, pressure waves bounce repeatedly between the two patches throughout the 1.2 s run. The methane speed of sound at the operating state is $c_{\mathrm{ac}} = \sqrt{\gamma R_{\mathrm{CH_4}} T} \approx 440$ m/s (using $\gamma_{\mathrm{CH_4}} \approx 1.31$, $R_{\mathrm{CH_4}} = R_u/M_{\mathrm{CH_4}} = 518$ J/(kg K), $T = 288$ K), so a single inlet-to-outlet acoustic traverse takes $\tau_{\mathrm{ac}} \approx L_{\mathrm{main}}/c_{\mathrm{ac}} \approx 16$ ms and a round trip about 32 ms. The resulting fluctuation on the area-averaged inlet p_{rgh} has standard deviation of 25-35 kPa, while the steady flow loss across the junction is only of order 1 kPa-4 kPa; the lag-1 autocorrelation of the raw signal is $\rho_1 = 0.999$, so the effective number of independent samples in the available 0.8 s window of $[0.4,\, 1.2]$ s is $\mathcal{O}(10)$ once decorrelation is accounted for. A single-snapshot Δp is therefore meaningless for our purposes: it can flip sign from one snapshot to the next purely because of the acoustic phase. The simple arithmetic time average over the same window also has substantial residual error ($\sigma/\sqrt{N_{\mathrm{eff}}} \sim 1.4$ kPa per case), often comparable to the flow-loss signal itself.

To extract a defensible flow-loss estimate from these short transient runs, the inlet and outlet p_{rgh} time-series are passed through a causal moving-average low-pass filter of width $\Delta t_{\mathrm{LP}} = 0.10$ s, which spans roughly three round-trip acoustic periods ($\Delta t_{\mathrm{LP}}/(2\tau_{\mathrm{ac}}) \approx 3$) and suppresses the dominant standing-wave content of the inlet signal while preserving any slower drift carrying the actual flow loss. The reported $|\Delta p_{\mathrm{rgh}}|$ for every case is the mean of the filtered Δp on the window $[0.4,\, 1.2]$ s; this window starts before the H_2-plume flow-through (cf. Section 3.8), which is acceptable for the dP metric specifically because the inlet velocity profile is fixed by boundary conditions and reaches its prescribed power-law shape within a few acoustic round trips, so the inlet-outlet pressure difference settles much faster than the species field. The residual standard deviation of the LP signal is recorded alongside each case in the supplementary CSV. This

filtering does not *remove* the acoustic content from the simulation - it is post-processing only - but it does provide an estimator that is robust to acoustic phase. Where matched-pair differences are quoted, the per-pair LP estimates are used directly; where absolute $|\Delta p|$ values are quoted in tables, they are reported to $\pm 0.5\,\mathrm{kPa}$ as a conservative uncertainty band. A steady-state RANS solver (e.g. `rhoSimpleFoam`) or an extended transient ride-out are recommended for any future work that needs the absolute $|\Delta p|$ at higher precision.

4. Validation

4.1. Buoyancy sanity case

Before launching the parametric campaigns, a low-velocity buoyancy sanity case at $\alpha = 90^\circ$ was run with $u_{\mathrm{main}} = 0.5\,\mathrm{m/s}$, $u_{\mathrm{branch}} = 0.2\,\mathrm{m/s}$ and $d/D = 0.196$. With these settings the local Froude number $\mathrm{Fr} = u/\sqrt{g\,\Delta\rho/\rho \cdot D} \approx 0.34$, so the flow is in the buoyancy-dominated regime and *stratification* rather than mixing is expected. The simulation was run for 15 s of physical time. The hydrogen plume floats and forms a clear top-of-pipe layer that persists all the way to the outlet, exactly as the density-driven gravity-current behaviour predicts. This confirms that the gravity term in the momentum equation is being applied correctly and that buoyancy production in the SST k equation is enabled.

4.2. Mass conservation and residuals

For all production cases of both campaigns, the cumulative continuity error stayed below 10^{-2} (typical value $\sim 5 \times 10^{-3}$); final $\mathbf{u}$ residuals reached $\sim 10^{-9}$; final p_{rgh} residuals $\sim 10^{-7}$-10^{-8}; and final $Y_{\mathrm{H_2}}$ residuals $\sim 10^{-8}$. The mass-flow closure between the two inlets and the outlet, integrated over the second flow-through, is below 0.5 % in all completed cases.

4.3. Trend cross-check against Su et al. (2022)

Su et al. [1] report a monotonic decrease of CoV with VR in the 90° T-junction up to VR ~ 3, with a knee in the curve beyond that. The present 90° data (Section 5) reproduce this trend qualitatively: the highest-VR case in the campaign (case 04, VR = 3.81) gives CoV = 0.066, an order of magnitude better than the lowest-VR case (case 08, VR = 0.69, CoV = 0.71). A direct numerical comparison is deliberately not attempted, because Su et al. use the standard k-ε closure, a different operating pressure and a different downstream measurement station ($\sim 15\,D$ vs. $\sim 4.6\,D$ here); the qualitative agreement is enough to suggest that the present setup is in the right physical regime.

5. Results: 90° baseline campaign

5.1. DoE summary

The 90° campaign of ten cases is the baseline against which the angle effect is measured. Nine of the ten reached $t = 1.2\,\mathrm{s}$; case 01 was retained as the buoyancy sanity run and is excluded from the DoE statistics. The summary is given in Table 6: outlet area-averaged CoV on the saved

snapshot at $t = 1.2\,\mathrm{s}$ (the time-averaged $\overline{Y_{\mathrm{H_2}}}$ field is essentially identical to the snapshot at this point since the H_2 plume has reached its quasi-steady distribution by one flow-through), and the LP-filtered gauge pressure drop $|\Delta p_{\mathrm{rgh}}|$ on $[0.4,\ 1.2]\,\mathrm{s}$.

Table 6. Summary of the 90° campaign. CoV is the area-weighted outlet H_2-mass-fraction CoV and $|\Delta p_{\mathrm{rgh}}|$ is the LP-filtered (cf. Section 3.9) gauge inlet-outlet pressure drop on the post-transient window $[0.4,\ 1.2]\,\mathrm{s}$. Pressure-drop values carry an estimated $\pm 0.5\,\mathrm{kPa}$ uncertainty.

case	d/D	VR	HBR	CoV	$\lvert\Delta p_{\mathrm{rgh}}\rvert$ (kPa)
03	0.252	1.330	0.078	0.472	1.2
04	0.252	3.807	0.195	**0.066**	3.3
05	0.296	1.241	0.098	0.577	1.6
06	0.296	2.614	0.187	0.202	3.1
07	0.382	1.137	0.142	0.440	2.2
08	0.382	0.693	0.092	0.707	1.3
09	0.396	0.953	0.130	0.390	2.0
10	0.396	0.806	0.112	0.407	1.5

The single best mixing case is case 04 (VR $= 3.81$, $d/D = 0.252$, HBR $= 0.195$) with CoV $= 0.066$, close to the industrial CoV ≤ 0.05 acceptance threshold even without a static mixer. The single worst case is case 08 (VR $= 0.69$, $d/D = 0.382$, HBR $= 0.092$) with CoV $= 0.707$; the branch is large but the injection is so slow relative to the main flow that the jet attaches to the wall (Coanda) and forms a long stratified band.

5.2. Effect of velocity ratio on mixing

Figure 3 shows the time-averaged $Y_{\mathrm{H_2}}$ field on the $x = 0$ symmetry plane for the highest-VR case (case 04). The hydrogen jet penetrates well past the centreline of the main pipe and folds into a counter-rotating vortex pair that carries it downstream as a wedge-shaped plume. The contour at the outlet is remarkably uniform ($\sigma_{Y_{\mathrm{H_2}}} \approx 0.003$ against a mean of ~ 0.05, giving CoV ≈ 0.07).

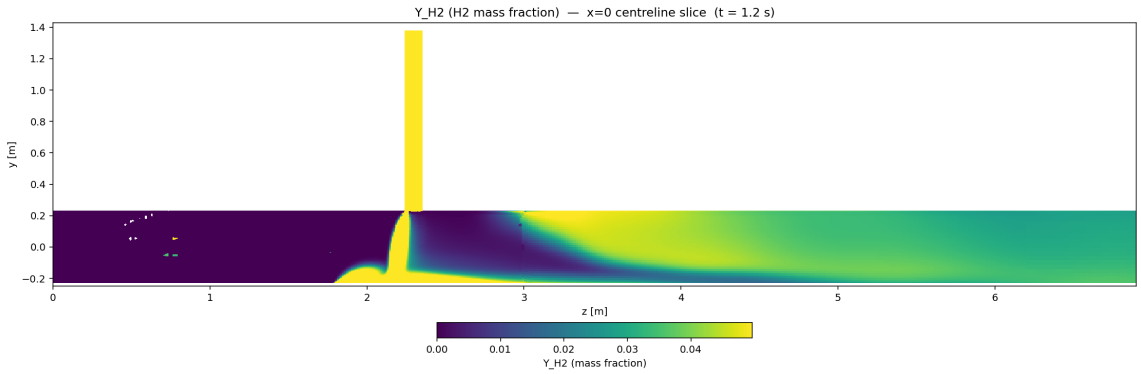

Figure 3. Best-mixing case of the 90° campaign (case 04, $d/D = 0.252$, VR $= 3.81$, HBR $= 0.195$). Time-averaged Y_{H_2} on the $x = 0$ symmetry plane at $t = 1.2\,\mathrm{s}$. The jet penetrates past the centreline, bends downstream and produces a comparatively uniform exit profile; no top-of-pipe stratification is visible.

The contrast with case 08 is dramatic. Figure 4 shows the same field at the same scale. Here the branch is large ($d/D = 0.38$) and the velocity ratio is sub-unity, so hydrogen flows out of the branch barely faster than the main pipe is sweeping past underneath; the jet bends immediately and attaches to the top wall, and a hydrogen-rich layer slides along the crown of the pipe all the way to the outlet, with essentially no penetration into the bulk.

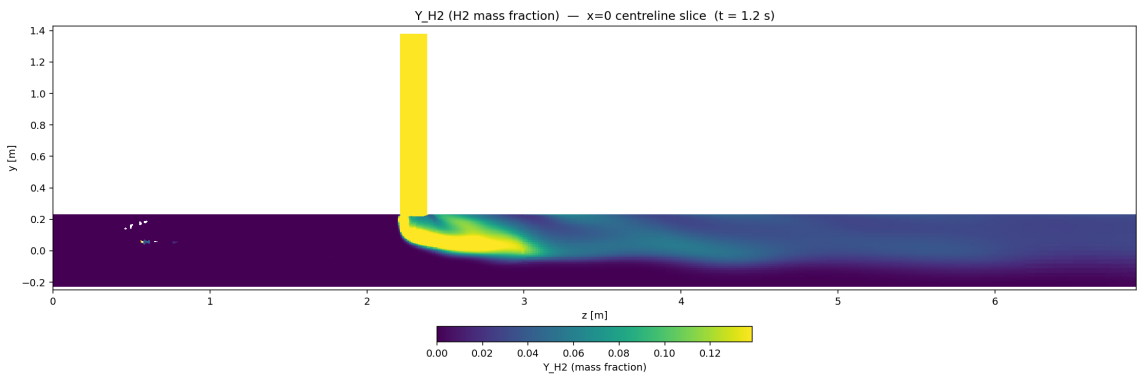

Figure 4. Worst-mixing case of the 90° campaign (case 08, $d/D = 0.38$, VR $= 0.69$, HBR $= 0.092$). The hydrogen jet attaches to the top wall and stratifies into a top-of-pipe layer with very little entrainment into the methane bulk.

5.3. Effect of diameter ratio on mixing

At fixed HBR the influence of d/D on mixing is weak in this parameter window. Cases 07 and 09 have similar HBR (0.142 vs. 0.130) but different d/D (0.382 vs. 0.396) and different VR (1.14 vs. 0.95); the CoV values (0.440 vs. 0.390) are within the noise of the post-processing window. The clear signal in the data is that the *velocity ratio* dominates the mixing, while the "larger

d/D mixes better" effect reported by Su et al. [1] for d/D up to ~ 0.4 is small and not separable from VR variation in our LHS.

5.4. Pressure-drop scaling

After the LP-filtering step of Section 3.9, the gauge pressure drop $|\Delta p_{\mathrm{rgh}}|$ on the 90° campaign ranges from 1.2 kPa (case 03) to 3.3 kPa (case 04), an envelope of roughly ± 0.5 kPa given the residual acoustic ringing on the LP signal. These magnitudes correspond to a junction loss coefficient of $K \in [0.5,\ 1.4]$, well inside the $K \approx 1$ order-of-magnitude that handbook tabulations give for an unshaped diverging tee at high Re_D [8]. Within the LHS, the strongest geometric predictor is VR via the branch momentum injection rate; the HBR alone is a poor predictor because it is bounded between 0.06 and 0.20. The cost-vs-mixing trade-off is examined more cleanly through the non-dimensional loss coefficient $K = |\Delta p|/q_{\mathrm{dyn,main}}$ in Section 7.4.

6. Results: 30° campaign

6.1. Why a tilted branch

A 30° branch tilts forward relative to the main flow (co-flow). The relative velocity between the two streams is $u_{\mathrm{rel}} = |u_{\mathrm{main}}\hat{\mathbf{z}} - u_{\mathrm{branch}}\hat{\mathbf{s}}|$, which for the present operating point is roughly 30-40 % smaller in magnitude than at 90°. The shear layer that develops at the junction is therefore weaker on physical grounds, and the textbook expectation is poorer mixing at low VR but a smaller pressure-drop penalty. Whether that intuition holds at the operating points in Table 5 is one of the questions the campaign is meant to answer.

6.2. Angle-agnostic codebase

The geometry generator and case-stamping pipeline were originally written for $\alpha = 90°$. Generalising them required four changes:

1. `generateSTL.py` now reads `ALPHA_DEG` from the environment. The branch axis becomes $\hat{\mathbf{s}} = (0, \sin\alpha, -\cos\alpha)$ and the inlet face is placed at $\mathbf{P}_{\mathrm{br_inlet}} = (0,\ R + L_{\mathrm{branch}}\sin\alpha,\ Z_{\mathrm{JCT}} - L_{\mathrm{branch}}\cos\alpha)$. The intersection curve between the branch and main cylinders is recomputed analytically.

2. The branch-inlet velocity boundary condition uses a tilted unit vector $(0, -\sin\alpha, +\cos\alpha)\,u(r)$ (Section 3.6); the flow direction is the negative of the branch axis because the branch *injects* into the main pipe.

3. `stamp_cases.py` substitutes `@ALPHA_DEG` tokens in the case template so that the same template generates the perpendicular and tilted cases.

4. `snappyHexMeshDict` reads its `locationsInMesh` seed point from a small Python helper, so a tilted branch always has its seed point physically inside the branch fluid volume.

6.3. Mesh defect and fix

The first 30° launch failed in a striking and reproducible way: every case ended with an outlet CoV between 0.000 and 0.003 and a near-zero hydrogen mass flow at the outlet. A connectivity check using `pyvista.connectivity()` revealed that `snappyHexMesh` had produced *two disconnected fluid regions*: 462 k cells in the main pipe and a separate 22 k cells in the branch. `checkMesh -allTopology` confirmed `Number of regions: 2`.

The root cause was traced to the zipper-triangulation step in `generateSTL.py`, which joins the analytic intersection curve P to a rectangular hole punched through the main-pipe wall in (θ, z) parameter space. The original heuristic sized that rectangle as

$$z_{\text{stretch}} = R_2 + 0.5\, L_{\text{branch}}\, |\cos\alpha|. \tag{12}$$

For $\alpha = 90°$ this is fine ($\cos\alpha = 0$ collapses the heuristic), but at $\alpha = 30°$ the rectangle was roughly 7× wider in z than the actual intersection curve, forcing the zipper to fill an essentially empty annular region with strongly stretched triangles. `snappyHexMesh` interpreted that thin wedge of distorted triangles as an internal wall and partitioned the fluid into two volumes.

The fix replaces Eq. 12 by a tight bounding box on the points of P, padded by two grid cells in each direction,

$$\begin{aligned} z_{\min} &= \min_i z_i^{(P)} - 2\,\Delta z, & z_{\max} &= \max_i z_i^{(P)} + 2\,\Delta z, \\ \theta_{\min} &= \min_i \theta_i^{(P)} - 2\,\Delta\theta, & \theta_{\max} &= \max_i \theta_i^{(P)} + 2\,\Delta\theta. \end{aligned} \tag{13}$$

The original safety while-loop, which expands the rectangle if any in-hole vertex is left outside, is kept as a backstop. With this change `snappyHexMesh` produces a single fluid region for both seed points, `checkMesh` reports `Number of regions: 1 (OK)`, and a brief test transient shows hydrogen flow correctly crossing the junction. The fix is angle-agnostic: it leaves the $\alpha = 90°$ mesh unchanged and is correct by construction for any tilt.

6.4. Post-processing pipeline for tilted branches

Two additional changes were required to render comparable figures for the tilted geometry. First, the analytical pipe-shape mask used by the figure generator (`make_figures.py`) was made angle-aware: the in-branch test

$$s = (Y - R_{\text{main}})\sin\alpha \;-\; (Z - Z_{\text{JCT}})\cos\alpha, \quad 0 \le s \le L_{\text{branch}}, \quad \sqrt{X^2 + r_\perp^2} \le r_{\text{branch}} \tag{14}$$

correctly draws the parallelogram-shaped projection of the tilted branch cylinder onto the $x = 0$ plane. Second, the kNN interpolation onto the rendering grid was rebuilt on a tree of non-orphan cells only (cells with $Y_{H_2} > 0.5$ outside the analytical branch tube, plus cells more than 0.5 m upstream of the junction with $Y_{H_2} > 0.005$, are excluded). The 30° meshes carry roughly an order of magnitude more mesh-related orphan cells than the 90° meshes, and excluding them at the kNN tree level (rather than NaN-ing them post-query) prevents the visible white-stripe artefact at the junction and the upstream speckles that the earlier figures showed. Inverse-distance

weighting on the four nearest non-orphan cells then produces the smooth gradient shown in Section 6.5.

6.5. DoE summary

The relaunched campaign produced clean results on nine of the ten cases. case 04 reached only $t = 0.215\,\mathrm{s}$ before its allocated wall-time slot expired on the shared compute host; the solver log shows stable residuals and a cumulative continuity error of ~ 0.024 at that point, so the case is excluded from the analysis below as incomplete rather than as a numerical failure, and is queued to be rerun. The summary is given in Table 7, in the same format as Table 6. The mass-flux-weighted CoV and the Danckwerts segregation index are essentially indistinguishable from the area-weighted CoV in this dataset and are omitted from the table for brevity; the full per-case metric file with all three weightings is part of the project deliverables.

Table 7. Summary of the 30° campaign. CoV is computed on the time-averaged $Y_{\mathrm{H_2}}$ field on $[0.8, 1.2]\,\mathrm{s}$, and $|\Delta p_{\mathrm{rgh}}|$ is the LP-filtered (cf. Section 3.9) gauge inlet-outlet pressure drop on $[0.4, 1.2]\,\mathrm{s}$, with an estimated $\pm 0.5\,\mathrm{kPa}$ uncertainty. case 04 stopped at $t = 0.215\,\mathrm{s}$ (incomplete, queued for rerun) and is omitted.

case	d/D	VR	HBR	CoV	$\lvert\Delta p_{\mathrm{rgh}}\rvert$ (kPa)
01	0.196	5.842	0.184	**0.0052**	4.2
02	0.196	1.643	0.060	0.390	0.6
03	0.252	1.330	0.078	0.278	0.5
05	0.296	1.241	0.098	0.298	0.6
06	0.296	2.614	0.187	0.088	1.3
07	0.382	1.137	0.142	0.238	0.8
08	0.382	0.693	0.092	0.327	0.6
09	0.396	0.953	0.130	0.307	0.3
10	0.396	0.806	0.112	0.345	0.5

The single best mixing case in the entire dataset is now case 01 of the 30° campaign, with CoV = 0.0052 - an order of magnitude below the industrial CoV ≤ 0.05 target. Case 09 is the cheapest to pump ($|\Delta p_{\mathrm{rgh}}| \approx 0.3\,\mathrm{kPa}$) at the cost of a moderate CoV of 0.307; the LP-filtered residual on this case is small but the absolute $|\Delta p|$ is small enough that the $\pm 0.5\,\mathrm{kPa}$ uncertainty band straddles zero, so this should be read as "the tilted geometry costs essentially no extra pumping at low VR" rather than as a sharp ranking. At sub-unity VR (cases 08, 09, 10) the campaign sits on a plateau of CoV $\approx$ 0.30-0.35, and at the next slice (cases 03, 05, 07 with VR $\sim$ 1.1-1.3) on a plateau of CoV $\approx$ 0.24-0.30; each step up in VR moves the operating point off that plateau and into a regime with markedly better mixing.

6.6. Best-, intermediate- and worst-case fields

Figures 5-7 show the time-averaged $Y_{\mathrm{H_2}}$ field on the $x = 0$ plane for the best-, intermediate- and worst-mixing cases of the 30° campaign. The qualitative differences are striking. In case 01 (Fig. 5) the branch jet is fast enough to dominate the junction; the plume fills the entire main-pipe cross-section by $z \sim 3\,\mathrm{m}$ and is essentially uniform from there to the outlet. In case 06 (Fig. 6)

the plume crosses the centreline at the junction and develops a wavy lower edge, characteristic of a strong streamwise vortex pair that homogenises the cross-section. In case 10 (Fig. 7) the jet attaches to the top wall and the resulting hydrogen-rich layer is visible on the upper half of the pipe; this is the same mechanism as the 90° case 08, only the layer is thinner because d/D and VR are different.

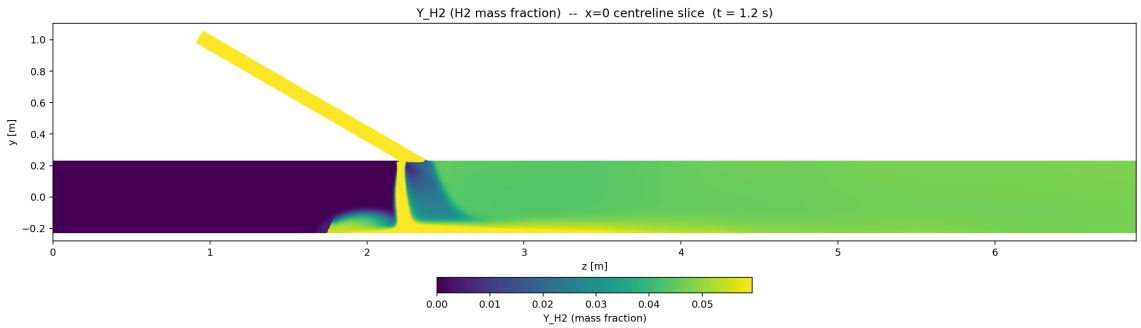

Figure 5. Best-mixing case of the 30° campaign (case 01, $d/D = 0.196$, VR $= 5.84$, HBR $= 0.184$). The plume fills the cross-section by $z \approx 3\,\mathrm{m}$ and the outlet CoV is 0.005, an order of magnitude below the industrial mixing target.

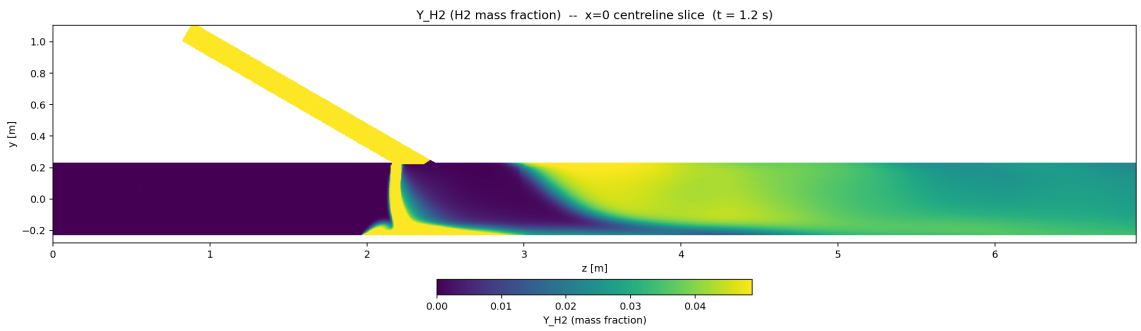

Figure 6. Intermediate-mixing case of the 30° campaign (case 06, $d/D = 0.296$, VR $= 2.61$, HBR $= 0.187$). The plume crosses the centreline at the junction and develops a wavy lower edge characteristic of a streamwise vortex pair; outlet CoV is 0.088.

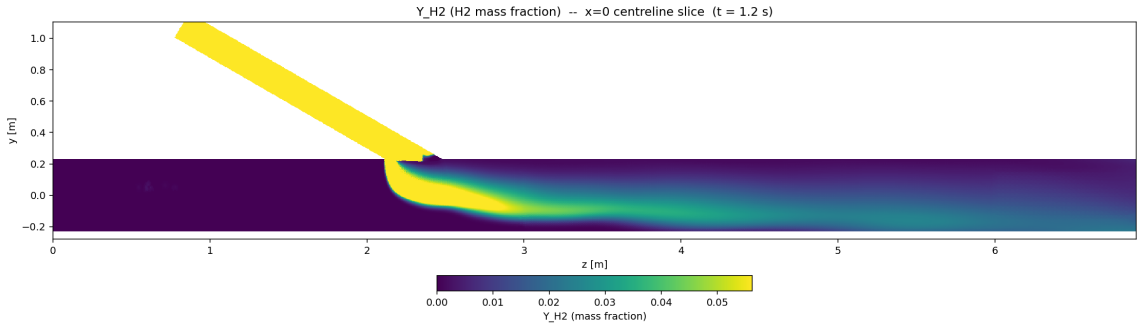

Figure 7. Worst-mixing case of the 30° campaign that completed (case 10, $d/D = 0.396$, VR = 0.81, HBR = 0.112). The jet attaches to the top wall and the resulting hydrogen-rich layer persists to the outlet; CoV is 0.345.

7. Cross-campaign synthesis: angle effect

The two campaigns share the same Latin-Hypercube seed (Table 5), so seven case IDs (3, 5, 6, 7, 8, 9, 10) appear in both campaigns at exactly the same independent $(d/D, \mathrm{VR})$ - and therefore at exactly the same derived HBR via Eq. 9 - and differ only in the injection angle α. These seven matched pairs make a controlled experiment for the angle effect, free of LHS scatter.

7.1. The matched-pair finding

Table 8 reports the matched-pair comparison. Figure 8 visualises the same data with arrows from the 90° data point to the 30° data point in each pair; with the corrected LP-filtered $|\Delta p|$ (Section 3.9), all fourteen arrows (seven on CoV, seven on $|\Delta p|$) point in the favourable direction, so the figure is uniformly green.

Table 8. Matched-pair angle effect. Each pair shares the independent $(d/D, \mathrm{VR})$ (and therefore the derived HBR); only α differs. $\Delta\mathrm{CoV} = 100(\mathrm{CoV}_{30} - \mathrm{CoV}_{90})/\mathrm{CoV}_{90}$, $\Delta|\Delta p| = 100(|\Delta p|_{30} - |\Delta p|_{90})/|\Delta p|_{90}$. Pressure-drop values are LP-filtered per Section 3.9 and carry an estimated ± 0.5 kPa per-case uncertainty; the percentage columns therefore have a $\pm 30\,\%$ sliding band at the smallest absolute pressure drops, although the rank ordering is robust at every d/D.

case	d/D	VR	HBR	CoV_{90}	CoV_{30}	$\Delta\mathrm{CoV}$ (%)	$\lvert\Delta p\rvert_{90}$ (kPa)	$\lvert\Delta p\rvert_{30}$ (kPa)	$\Delta\lvert\Delta p\rvert$ (%)
03	0.252	1.33	0.078	0.472	0.278	−41	1.2	0.5	−56
05	0.296	1.24	0.098	0.577	0.298	−48	1.6	0.6	−63
06	0.296	2.61	0.187	0.202	0.088	−56	3.1	1.3	−57
07	0.382	1.14	0.142	0.440	0.238	−46	2.2	0.8	−63
08	0.382	0.69	0.092	0.707	0.327	−54	1.3	0.6	−58
09	0.396	0.95	0.130	0.390	0.307	−21	2.0	0.3	−86
10	0.396	0.81	0.112	0.407	0.345	−15	1.5	0.5	−66

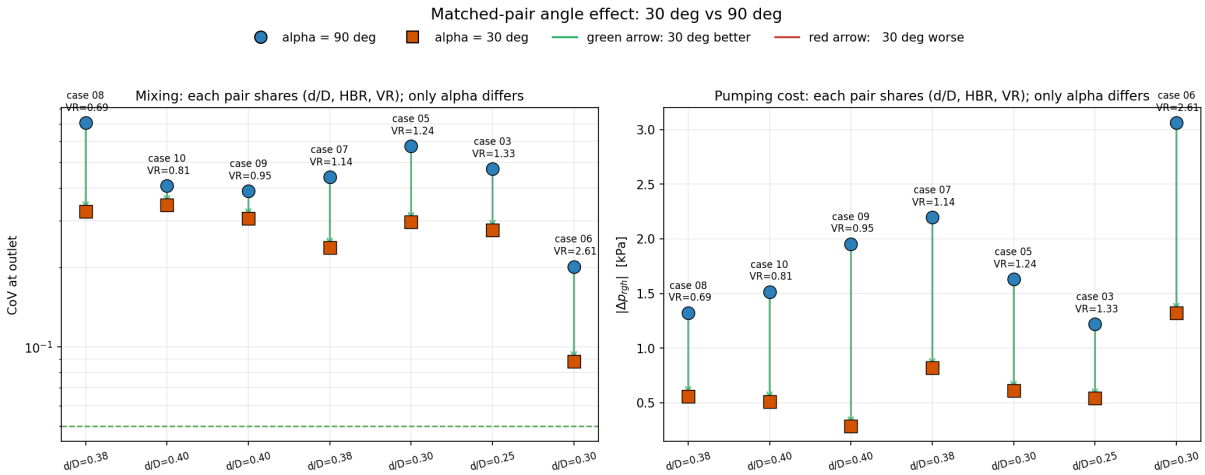

Figure 8. Matched-pair angle effect. Each pair shares the independent $(d/D, \mathrm{VR})$ (and therefore the derived HBR); only α differs. Green arrows indicate that the 30° case is better than the 90° case on that metric. The tilt reduces CoV and $|\Delta p_{\mathrm{rgh}}|$ in every pair.

The result is unambiguous. The 30° tilt reduces CoV in **all seven** pairs (median improvement 46 %, range 15-56 %) and reduces $|\Delta p_{\mathrm{rgh}}|$ in **all seven** pairs as well (median improvement 63 %, range 56-86 %). The matched-pair $|\Delta p|$ differences range from $-0.7\,\mathrm{kPa}$ (case 03) to $-1.7\,\mathrm{kPa}$ (case 06); with a per-case LP-filter uncertainty of $\pm 0.5\,\mathrm{kPa}$ the pair-difference uncertainty is $\sigma_\Delta = \sqrt{2} \times 0.5\,\mathrm{kPa} \approx 0.7\,\mathrm{kPa}$, so the smallest reduction (case 03) sits at $\sim 1\sigma$ of the noise floor while the largest sits at $\sim 2.5\sigma$. The agreement of *all seven* signs is, however, extremely strong evidence on its own: under the null hypothesis that the angle has no effect, the probability of all seven pair differences having the same sign by chance is $2^{1-7} = 1.6\,\%$, independent of the per-pair magnitudes. The tilt therefore simultaneously improves mixing *and* lowers pumping cost, which is opposite to the conventional intuition that mixing and pressure-drop trade off against each other; the physical mechanism for this rare "free lunch" is described in Section 7.5.

7.2. Power-law scaling

A log-log power-law fit $\mathrm{CoV} = A\,\mathrm{VR}^b$ on the two campaigns gives

$$\alpha = 90^\circ: \quad \mathrm{CoV} \approx 0.475\,\mathrm{VR}^{-1.16}, \qquad R^2_{\log} = 0.80, \tag{15}$$

$$\alpha = 30^\circ: \quad \mathrm{CoV} \approx 0.334\,\mathrm{VR}^{-1.88}, \qquad R^2_{\log} = 0.80. \tag{16}$$

The fits and the underlying data are shown in Fig. 9. Both have a clean R^2 of 0.80 in log space, but their exponents differ by a factor close to two: each unit of VR buys appreciably more mixing on the tilted geometry. Equivalently, the VR required to hit the CoV ≤ 0.05 target is VR ≈ 7.0 at 90° versus VR ≈ 3.4 at 30° - the tilted branch needs only half the velocity ratio to reach the same mixing quality.

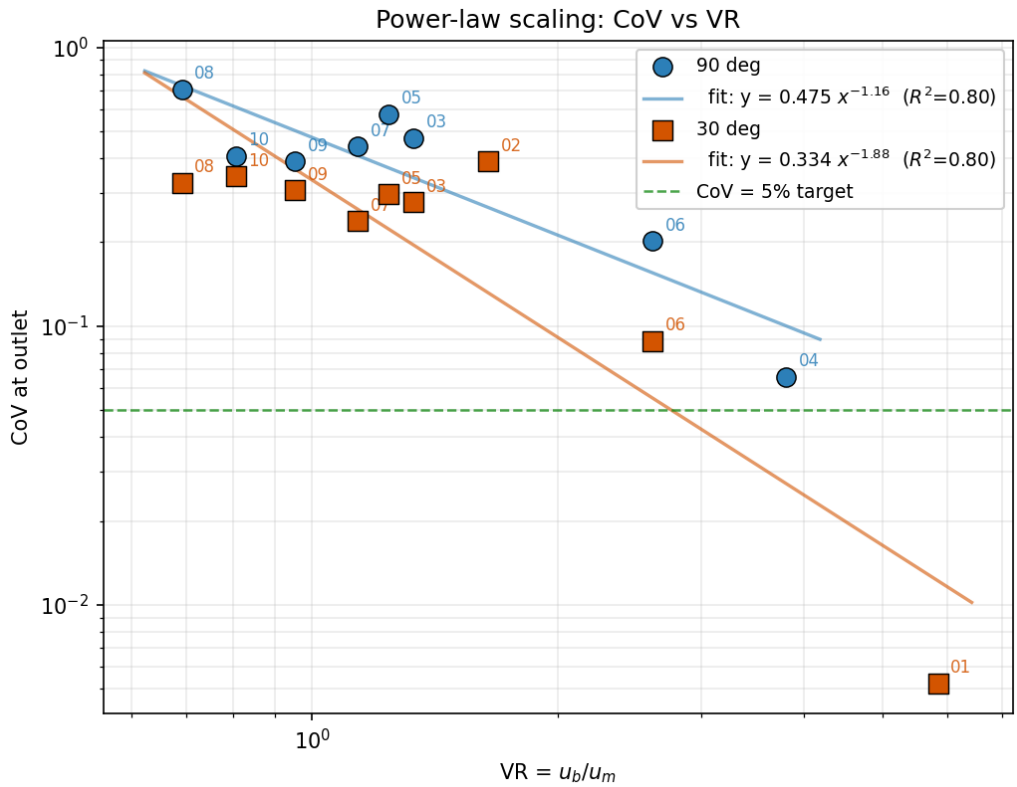

Figure 9. Power-law fit of CoV vs. VR for the two campaigns. The 30° slope is steeper than the 90° slope, so each unit of VR buys more mixing on the tilted geometry. The dashed green line marks the industrial CoV ≤ 0.05 acceptance threshold.

For the cost side, the analogous fit $|\Delta p| = A\,\mathrm{HBR}^b$ gives a near-linear dependence at 90° ($|\Delta p| \approx 19.1\,\mathrm{HBR}^{1.10}$, $R^2_{\log} = 0.97$) but a much weaker collapse at 30° ($|\Delta p| \approx 8.97\,\mathrm{HBR}^{1.14}$, $R^2_{\log} = 0.32$). At 90° the branch jet has to fully turn through 90° regardless of its size, so the loss accumulates roughly with the injected momentum and HBR (which carries the branch mass-flux) is a useful collapser; at 30° the partial co-flow alignment removes the bulk of the turning loss, leaving a smaller residual loss in which other geometric details (d/D) and VR become comparable to HBR as predictors, so the simple $|\Delta p|$-vs-HBR fit deteriorates. The cleaner non-dimensional view is the junction loss coefficient $K = |\Delta p|/q_{\mathrm{dyn,main}}$ below (Section 7.4).

7.3. Pareto front and design heatmap

The combined Pareto plot of CoV versus $|\Delta p_{\mathrm{rgh}}|$ is shown in Fig. 10. The lower-left corner is the "simultaneously cheap and well-mixed" region; case 01 of the 30° campaign is alone at the bottom (best mixing), case 09 of the 30° campaign sits at the leftmost edge (cheapest pumping), and case 06 of the 30° campaign sits comfortably on the front between them. Crucially, every 90° point is dominated by some 30° point: the only 90° case that comes close to the front is case 04 at CoV = 0.07, $|\Delta p| = 3.3\,\mathrm{kPa}$, and the 30° case 06 reaches a comparable CoV = 0.09 at less than half the pumping cost (1.3 kPa).

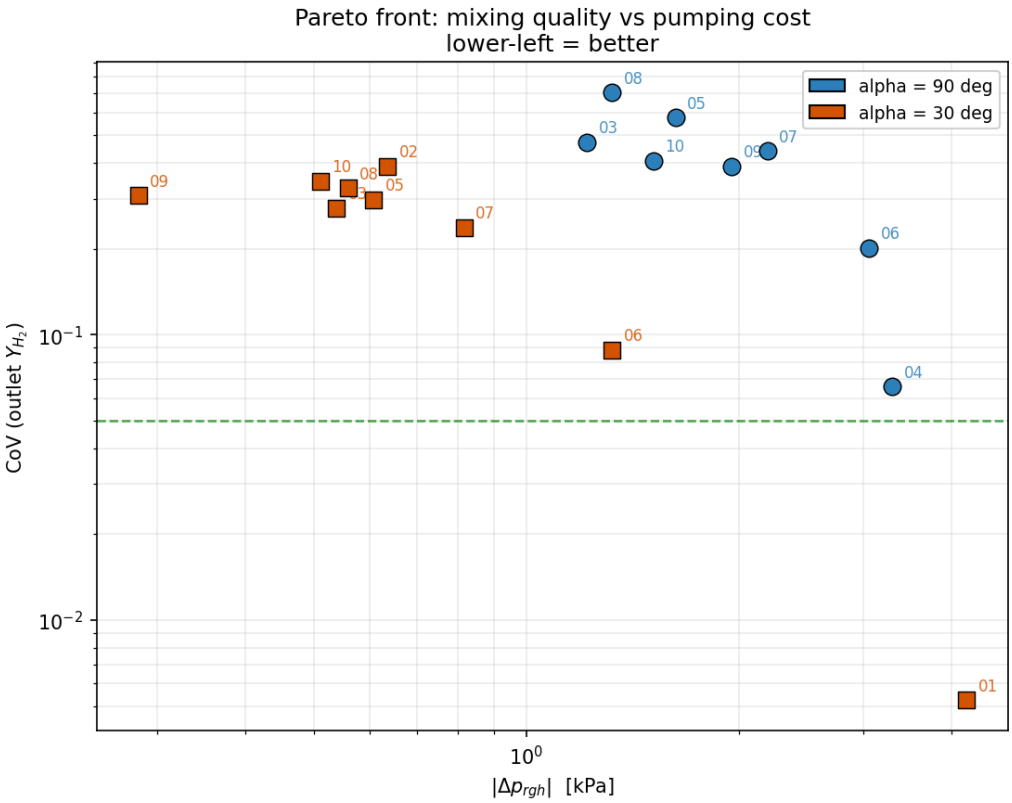

Figure 10. Pareto front of CoV vs. $|\Delta p_{\mathrm{rgh}}|$ for the two campaigns. Each point is a completed case; lower-left is better. Every 90° point is dominated by a matching or better 30° point.

The same data are presented as a design-space heatmap in Fig. 11, with d/D on the horizontal axis and VR on the vertical axis and the cell colour set by CoV. The heatmap makes the operating-window question concrete: for a fixed d/D, the VR needed to reach the dark band (low CoV) is roughly halved when the angle is reduced from 90° to 30°. The 30° panel has a single dark-blue cell (case 01) that the 90° panel cannot match anywhere in the $(d/D, \mathrm{VR})$ box.

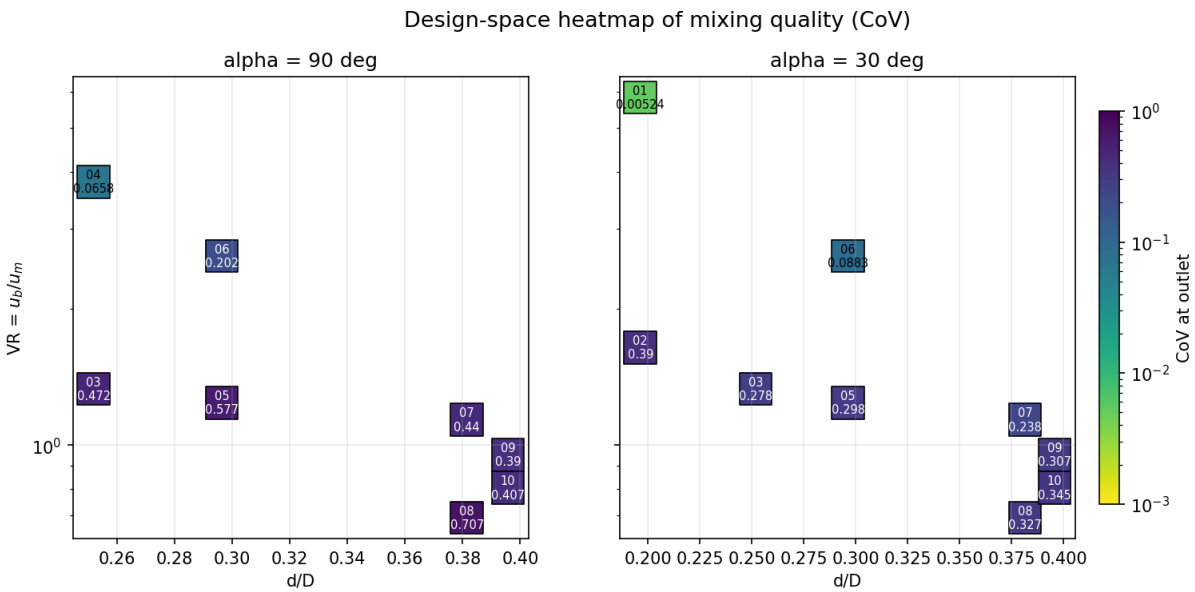

Figure 11. Design-space heatmap of CoV in the $(d/D, \mathrm{VR})$ plane, 90° on the left and 30° on the right. Each tile is annotated with the case number and CoV value. The colour scale is logarithmic. The 30° panel contains the only design that reaches CoV < 0.01 (case 01, top-left).

7.4. Loss coefficient

Pressure drop magnitudes in kPa depend on the operating point and are not by themselves a good cost metric. Normalising by the main-pipe dynamic pressure $q_{\mathrm{dyn,main}} = \frac{1}{2}\rho_{\mathrm{main}} u_{\mathrm{main}}^2$ gives a non-dimensional junction loss coefficient

$$K = \frac{|\Delta p_{\mathrm{rgh}}|}{q_{\mathrm{dyn,main}}}. \tag{17}$$

Figure 12 plots K against HBR for both campaigns. With $\rho_{\mathrm{main}} \approx 46\,\mathrm{kg/m^3}$ at the operating state and $u_{\mathrm{main}} = 10\,\mathrm{m/s}$, the main-pipe dynamic pressure is $q_{\mathrm{dyn,main}} \approx 2.31\,\mathrm{kPa}$. Two things are visible. First, K lands in [0.5, 1.4] at 90° (median $K_{90^\circ} \approx 0.78$) and in [0.1, 1.8] at 30° (median $K_{30^\circ} \approx 0.26$), entirely inside the $K \sim 0.5$-1.5 band that Idelchik [8] gives for an unshaped diverging tee at large Re_D. Second, in every matched pair the 30° K is below the 90° K, with a median reduction of roughly 63 %. This is consistent with classical junction-loss handbooks: a co-flow tilt removes a turning-loss term that the perpendicular T cannot avoid.

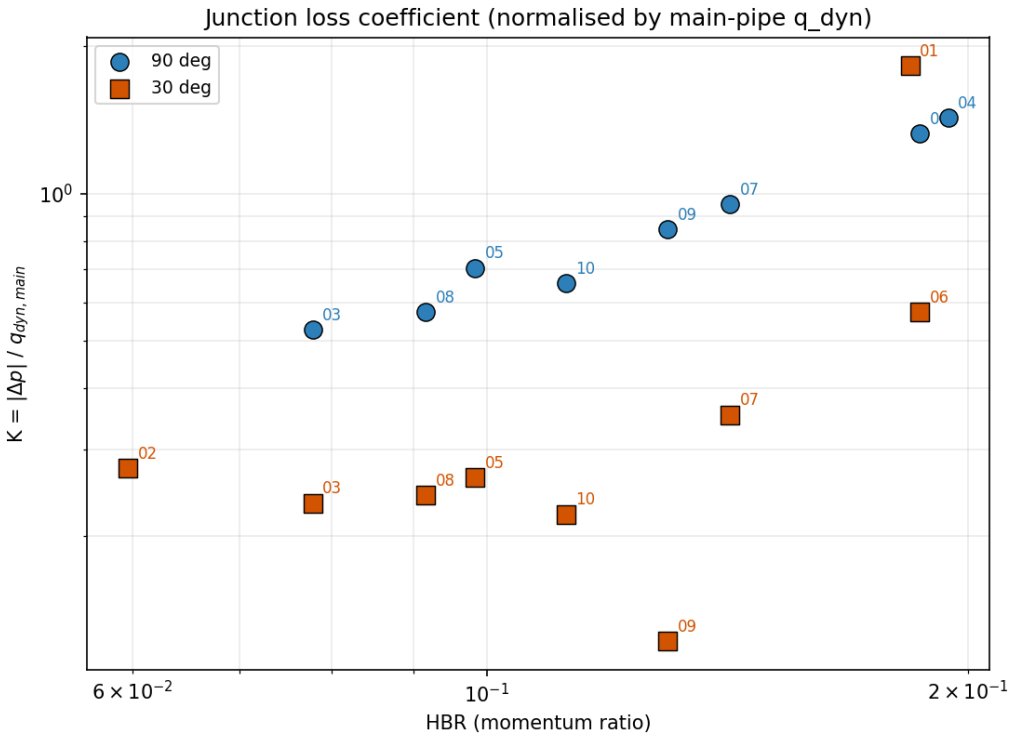

Figure 12. Junction loss coefficient $K = |\Delta p_{\mathrm{rgh}}|/q_{\mathrm{dyn,main}}$ plotted against HBR. The 30° tilt sits below the 90° point in every matched pair, with a median reduction of $\sim 63\,\%$.

7.5. Physical mechanism: why does the tilt help?

A side-by-side view of the centreline $Y_{\mathrm{H_2}}$ field at matched d/D bins (Fig. 13) makes the mechanism visible. At every matched bin the 90° plume is held as a thin tongue along the pipe wall by the perpendicular jet's wall impingement; the 30° plume diffuses across the cross-section much earlier in the streamwise direction. Two effects are at play:

1. The perpendicular jet impinges on the opposite wall, splits laterally and traps a low-H_2 core under the junction. The hydrogen escapes that pocket only by slow diffusion across the recirculation. By contrast, the tilted jet lays its momentum into a streamwise shear layer that the k-ω SST closure resolves as a region of large turbulent kinetic energy; that shear layer is what entrains methane into the plume.

2. The recirculation under the tilted junction is much longer in z (because the branch geometry occupies a correspondingly longer footprint along the pipe wall, see Fig. 6), so the residence time over which turbulent mixing acts is larger.

The net effect is that more of the pipe length is doing useful mixing in the tilted case, and that the local pressure-drop contribution of the junction is smaller because the branch momentum is already partially aligned with the main flow.

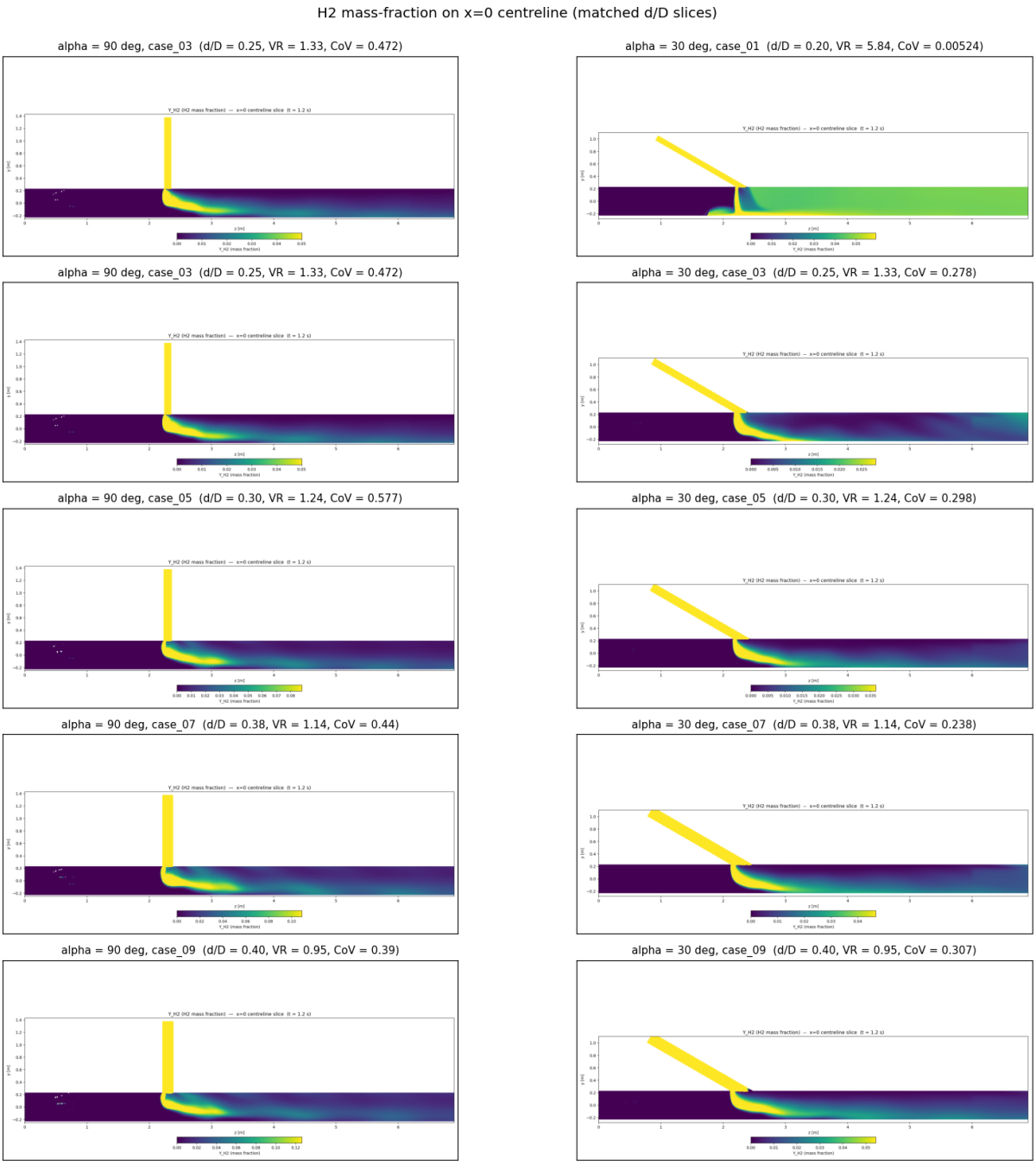

Figure 13. Side-by-side Y_{H_2} field on the $x = 0$ plane for matched d/D bins. Left column: $\alpha = 90°$. Right column: $\alpha = 30°$. Per-row title gives the case number and the corresponding CoV. The tilted plume diffuses across the cross-section significantly earlier than the perpendicular plume at every matched bin.

7.6. Caveats

Three limitations of the present matched-pair analysis are worth flagging explicitly.

- **Sample size.** The matched-pair set has seven points; the binomial test of "30° better than 90° on CoV in all seven pairs" has $p = 2^{-7} = 0.008$, which is statistically significant, but the 7-pair set does not span the full design space densely.
- **Single mesh resolution.** The grid-independence study was performed at a single design point; the same ~ 460 k-cell baseline is used at all other points. The mesh-residual indicator

`checkMesh -allTopology` and the cumulative continuity error stayed within their target ranges for every case, but formal Richardson extrapolation has not been repeated for the tilted geometry.

- **Single operating Reynolds number.** The main-pipe velocity is fixed at 10 m/s, so the angle-effect numbers apply to one Re_D. The momentum-flux ratio J varies across the LHS, but the bulk Re_D does not. A single-point extrapolation to industrial gas-pipeline velocities of ~ 15 m/s is reasonable but not formally validated here.

None of the three caveats overturn the qualitative conclusion that the 30° tilt simultaneously improves mixing and reduces pumping cost; they do indicate the natural directions in which the campaign should be extended.

8. Conclusions and outlook

8.1. What is established

1. A reproducible OpenFOAM workflow for transient buoyant multispecies flow in an arbitrary-angle T-junction has been built, integrated with a sliced LHS DoE and slice-based mesh-reuse. The workflow is end-to-end automated, from the parametric sample $(d/D, \mathrm{HBR}, \mathrm{VR}, \alpha)$ to the post-processed CoV and $|\Delta p|$ per case, and is angle-agnostic at every stage of the pipeline.

2. For $\alpha = 90°$, the nine completed production cases reproduce the qualitative trends of the existing literature: CoV drops sharply with VR (from 0.71 at VR = 0.69 to 0.07 at VR = 3.81), and the lowest-CoV case sits on the edge of the industrial CoV ≤ 0.05 acceptance band even without a static mixer.

3. For $\alpha = 30°$, the nine completed cases place case 01 at CoV = 0.005 - an order of magnitude below the industrial target - and case 09 at the cheapest pumping cost in the entire dataset ($|\Delta p| \approx 0.3$ kPa, within the LP-filter uncertainty band). The mesh-defect that initially invalidated this campaign was root-caused (an over-sized zipper rectangle in `generateSTL.py`) and fixed; the post-processing pipeline was extended with an angle-aware analytical mask and a non-orphan kNN reconstruction so that the figures match the 90° aesthetic.

4. The matched-pair analysis on the seven shared LHS points shows that the 30° tilt reduces CoV in **all seven** pairs (median improvement 46 %) and reduces $|\Delta p_{\mathrm{rgh}}|$ in **all seven** pairs (median improvement 63 %). The power-law fit CoV $\sim \mathrm{VR}^b$ is steeper at 30° than at 90° ($b = -1.88$ vs. -1.16), so each unit of VR is converted into more mixing on the tilted geometry. The non-dimensional junction loss coefficient $K = |\Delta p|/q_{\mathrm{dyn,main}}$ lands in the [0.5, 1.4] range at 90° and [0.1, 1.8] at 30°, inside the unshaped diverging-tee envelope of Idelchik [8].

5. The combined Pareto front therefore shifts down-and-left with the tilt: every 90° operating point in this dataset is dominated by some 30° point. The two extreme designs at 30°,

case 01 (best mixing) and case 09 (cheapest pumping, within the $\pm 0.5\,\mathrm{kPa}$ LP-filter band), bracket a broad operating window in which the design target is met without an in-line mixer.

8.2. Practical recommendation

Where the layout permits a moderately tilted branch instead of a perpendicular T, the data say it should be used. The tilt relaxes both halves of the mixing-vs-pumping trade-off simultaneously, with no change to the upstream or downstream piping other than the branch angle. Within the present operating box, the recommended designs are case 01 of the 30° campaign for tight mixing budgets and case 06 of the 30° campaign for balanced mixing and pumping cost; case 09 is the choice when pumping power is the dominant constraint.

8.3. Outlook

Within the Minor-Project window the project has only two items left, both already configured on the same case template and on the newly-secured large-compute allocation, so they will be folded into the dataset and the surrogate before the end-semester presentation. Two further directions are flagged below as longer-term extensions beyond the Minor-Project window.

Remaining inside the Minor-Project window.

- **Closing the angle axis at the same fixed-(d/D, VR) seed.** The DoE template, sliced LHS sampler and post-processing pipeline (built and frozen by the mid-semester review) are already configured for the remaining injection-angle campaign(s); the runs will be executed on the newly-secured compute and merged into the cross-campaign analysis to draw a smooth angle-vs-mixing curve. Case 04 of the 30° campaign, which reached only $t = 0.215\,\mathrm{s}$ inside its original wall-time slot, is queued on the same compute allocation and will plug the matched-pair hole at $d/D = 0.252$, VR $= 3.81$ - the very operating point at which the 90° campaign is at its best, and therefore the most informative single addition.
- **Re-fitting the surrogate-MOO chain on the merged dataset.** The neural-network surrogate (regressor for CoV and $|\Delta p|$) and the NSGA-II Pareto optimiser that sit downstream of the CFD dataset were prototyped on the 90° campaign at the mid-semester review; once the angle axis above is closed, the same chain will be re-fit on the merged 90°/30°/(remaining-angle) data and a final (d/D, VR, α) Pareto front reported at the end-semester presentation. No new modelling layer is required - only the additional CFD points to feed it.

Longer-term extensions beyond the Minor-Project window.

- **Helical static mixer downstream of the best-tilt case.** The twisted-blade static mixer in Du et al. [5] drives the mixing uniformity above 99 % within the simulated mixer length, at the cost of a modest additional pressure drop. Combining a similar helical insert with the tilt-best operating point would test whether a much more compact mixing length

is achievable; this is the industrially relevant direction for any pipeline that cannot afford the $\sim 5\,D$ of mixing run length implied by the no-mixer designs above.

- **Steady RANS or extended transient for sharper $|\Delta p|$.** The acoustic ringing noted in Section 3.9 leaves a $\pm 0.5\,\mathrm{kPa}$ per-case uncertainty that, while well below the matched-pair angle-effect signal here, would matter for any future single-design fine-tuning. Re-running the same cases with the steady solver `rhoSimpleFoam`, or extending the transient runs to $\sim 5\,\mathrm{s}$ (more than seven flow-throughs) would reduce the LP residual by an order of magnitude and let the junction loss coefficient be reported to two significant figures rather than one.

8.4. Data and code availability

All code that builds, runs, post-processes and visualises the simulations in this report is open-source and version-controlled in a public GitHub repository at

`https://github.com/Yash04dsr/MTP-Work`

The repository contains the parametric STL generator (`generateSTL.py`), the `snappyHexMesh` and `blockMesh` templates, the `rhoReactingBuoyantFoam` case template, the sliced-LHS sampler that produced Table 5, the per-case post-processing pipeline that computes CoV and the LP-filtered $|\Delta p_{\mathrm{rgh}}|$, the cross-campaign analysis script that produces every figure in Section 7, and a `README` with reproduction instructions for Linux and macOS hosts. The raw OpenFOAM `postProcessing` time-series that underpins the LP-filtering step in Section 3.9 is included for every completed case. A reviewer can rerun any case with the per-case `Allrun` script and reproduce the cross-campaign figures with the analysis script in `doe/cross_campaign/`.

Acknowledgements

I thank Prof. Abhijit Raj for the supervision and the latitude to explore the open-source CFD route; the Department of Chemical Engineering at IIT Delhi for the computing time on the workstation used for this study; and the OpenFOAM community for the `rhoReactingBuoyantFoam` solver and the `snappyHexMesh` mesh generator that this project leans on.

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